

Improving Micro-Expression Recognition through Temporal and Motion Modeling

Recognizing facial micro-expressions (MEs) [1] is a particularly difficult task because these expressions are extremely brief, subtle, and dynamic. Traditional approaches [2] based on hand-crafted descriptors often depend on manually selecting key frames, such as the onset or apex, while deep learning methods [3] require large and diverse datasets that are still scarce in this field. Both directions therefore face limitations, and recognition accuracy remains modest even on widely used datasets such as CASME-II, SAMM, CAS(ME)², and SMIC.

Two main factors explain these difficulties. First, accurate temporal localization is essential, since identifying the exact onset, apex, and offset frames of a micro-expression is critical for reliable classification. Even small errors in phase alignment can reduce recognition accuracy [4]. Second, micro-expressions involve very weak motion signals, consisting of subtle muscle movements that are easily confused with neutral expressions, background variations, or camera noise. Prior studies have attempted to address these challenges through motion magnification to amplify subtle changes [5], optical-flow-based motion descriptors [6], and spatio-temporal networks [7]. While these methods have produced incremental improvements, their impact remains limited, highlighting the need for a more systematic investigation of temporal and motion modeling strategies.

This thesis will investigate whether improved temporal modeling and motion representations can enhance recognition of micro-expressions in dynamic FER. The central research question is: how can modeling of temporal phases and motion cues improve recognition accuracy for subtle and short-lived expressions? The work will build on established supervised models and evaluate practical strategies such as temporal attention or hybrid motion representations. The expected outcome is a contribution that provides a clearer understanding of current limitations and demonstrates approaches for improving sensitivity to fine-grained dynamics while maintaining robustness to noise.

References

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